Project: Anti-Money Laundering (AML) Detection System

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Summary

Objective: Detect suspicious (money laundering) transactions using supervised and unsupervised approaches.

Data: SAML-D (cleaned/balanced), engineered features, and PCA variants.

Methods:

- Supervised: Baselines + advanced models (see notebooks).
- Unsupervised: K-means, DBSCAN, GMM (10 components from EDA).

Key Outcomes:

- Unsupervised methods identify anomalies without labels; GMM with optimized thresholds provides strong performance.
- Threshold optimization improves precision/recall trade-off.

Deliverables:

- Metrics table: [aml_project/data/processed/clustering_results.csv]
- Predictions: [aml_project/data/processed/clustering_predictions.csv]
- Figures: comparison, ROC curves, GMM visualizations and uncertainty plots.

1. Introduction

- Problem: Financial crime detection requires identifying rare, evolving laundering behaviors in transactional data.
- Goal: Build a robust pipeline for detecting AML using both supervised and unsupervised techniques, enabling anomaly discovery and operational thresholding.
- Notebooks:
    - EDA: [aml_project/notebooks/01_EDA.ipynb]
    - Preprocessing: [aml_project/notebooks/02_preprocessing.ipynb]
    - Feature Engineering: [aml_project/notebooks/03_feature_engineering.ipynb]
    - Models: [aml_project/notebooks/04_model_baseline.ipynb], [05_advanced_models.ipynb], [07_final_model.ipynb]
    - Unsupervised: [aml_project/notebooks/04c_clustering_predictions.ipynb]

2. Data Overview

- Source: SAML-D (CSV) with transactional attributes and Is_laundering label.
- Sizes:
    - Raw: see [aml_project/data/raw/SAML-D.csv]
    - Balanced sample (undersampled legitimate): [aml_project/data/raw/balanced_data.csv]
    - Processed features: [aml_project/data/processed/features.csv]
- Label Distribution:

    Number of features: 82

    Features shape: (58817, 82)

    Target shape: (58817,)

TRAIN-TEST SPLIT:

Training set: 47,053 samples (80.0%)

Test set: 11,764 samples (20.0%)

Training set fraud rate: 16.32%

Test set fraud rate: 16.32%

3. Preprocessing

- Steps:
  - Dropped Laundering_type for binary classification.
  - Balanced dataset to approximate a 5:1 legitimate-to-fraud ratio in EDA.
  - Encoded categorical columns.
  - Standardized features (StandardScaler) prior to modeling.
- Outputs:
  - Cleaned (EDA DBSCAN-based) optional file: [aml_project/data/raw/cleaned_data.csv] (if used further).
  - Processed feature matrix: [aml_project/data/processed/features.csv]

4. Feature Engineering

- Source: [aml_project/src/feature_engineering.py]
- Feature names: [aml_project/models/feature_names.txt]
- Optional PCA views also evaluated in EDA and clustering notebooks.

5. Modeling Approaches

- Supervised:
    - Baseline models: Logistic & Decision Tree
    - Advanced models: Random Forest & XGBoost
- Unsupervised :
    - Clustering predictions: Kmeans, PCA ,DBSCAN & GMM

6. Unsupervised Methods (Clustering-Based Anomaly Detection)

- Data Split: 80/20 stratified train/test on standardized features.
- Metrics on Test Set: Precision, Recall, F1; ROC-AUC where applicable.

6.1 K-means (k=2)

- Method 1: Cluster assignment (cluster with higher fraud rate = anomaly).
- Method 2: Distance-based scoring with F1-optimized threshold (from training).
- Outputs: Predictions (two ways), precision/recall/F1, ROC-AUC for distance-based.

6.2 DBSCAN

- Parameter selection via k-distance analysis and sample-based grid: eps and min_samples.
- Method 1: Noise points (-1) are anomalies.
- Method 2: High-fraud clusters (from training) also marked anomalous.
- Outputs: Predictions, metrics.

6.3 Gaussian Mixture Model (GMM)

- Optimal components from EDA: n=10 (BIC-based).
- Methods:
  - Cluster-based: cluster with highest fraud rate.
  - Score-based: percentile threshold (e.g., 90th) on anomaly scores.
  - Optimal threshold: training-based F1 optimization on anomaly scores.
- Outputs: Predictions, metrics, ROC-AUC; visualizations and uncertainty analysis.

7. Model Selection and Thresholding

- Threshold Optimization:
  - K-means distance and GMM score thresholds tuned on training via precision-recall F1.
- Best Method (from 04c):
  - BEST PERFORMING METHOD (on Test Set):

    K-means (Distance)

    F1-Score: 0.5486

    Precision: 0.4587

    Recall: 0.6823

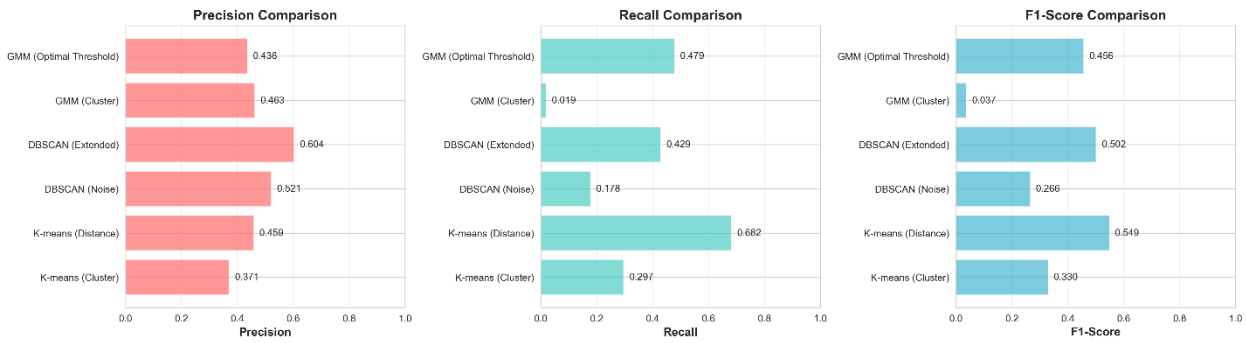
8. Results and Discussion

- Findings:
  - Unsupervised can surface novel patterns and complement supervised pipelines.
  - GMM with 10 components and F1-optimized threshold produced competitive balance of precision/recall.
  - DBSCAN flags density-based outliers effectively; K-means provides interpretable cluster-distance scores.
- Trade-offs:
  - Higher recall increases false positives; threshold tuning aligns with risk appetite.
  - GMM's soft clustering gives richer uncertainty/score signals than hard clustering.

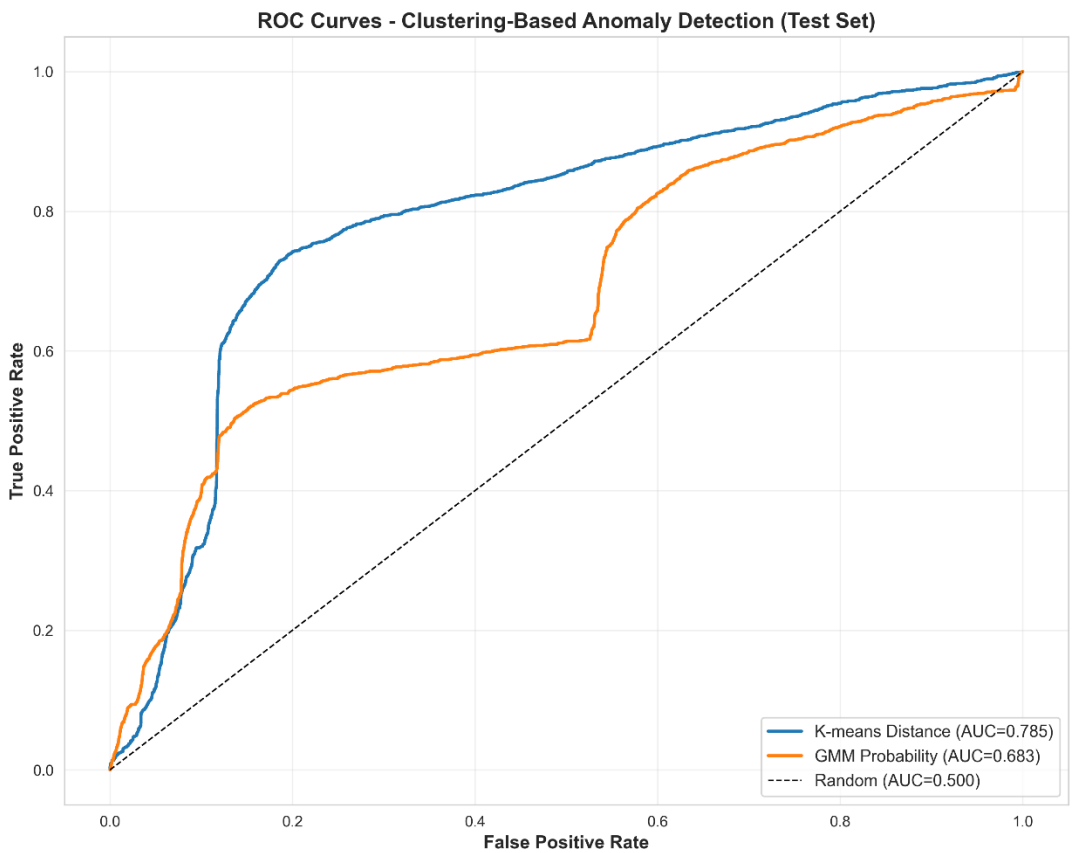
9. Reproducibility

- Environment: Python 3.x; install from [aml_project/requirements.txt].
- Run order:
  1. 01_EDA.ipynb (for EDA and GMM components = 10)
  2. 02_preprocessing.ipynb
  3. 03_feature_engineering.ipynb
  4. 04_model_baseline.ipynb
  5. 04b_imbalance_handling.ipynb
  6. 04c_clustering_predictions.ipynb
  7. 05_advanced_models.ipynb
  8. 05b_threshold_optimization.ipynb
  9. 06_model_comparison.ipynb
  10. 07_final_model.ipynb

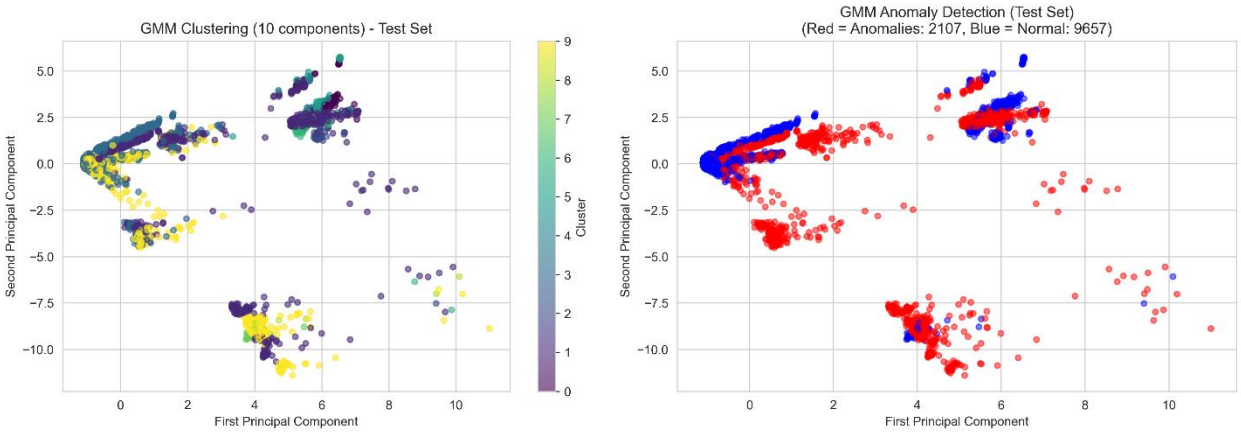
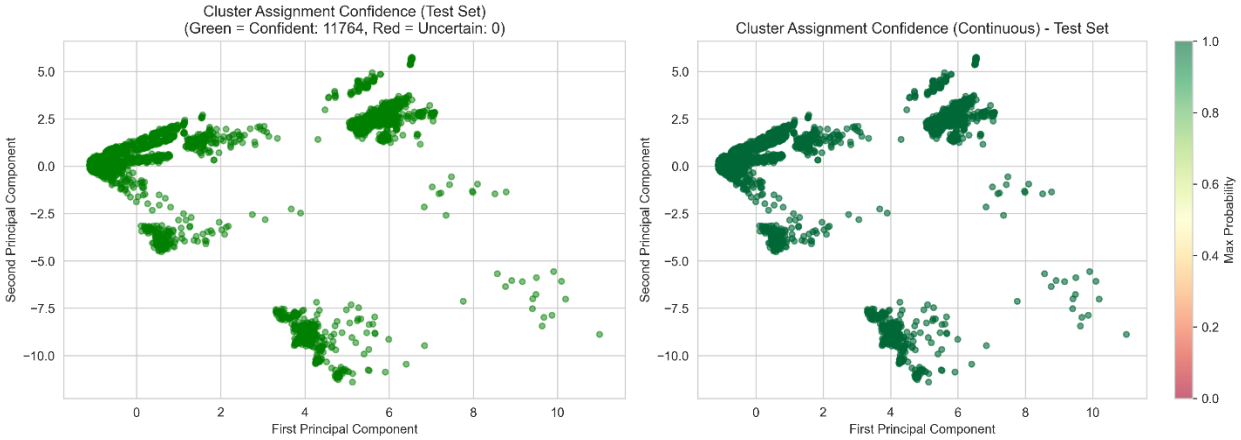
- Artifacts used in report:
    - [aml_project/data/processed/clustering_results.csv]
    - [aml_project/data/processed/clustering_predictions.csv]
    - [aml_project/data/processed/clustering_comparison.png]



- [aml_project/data/processed/clustering_roc_curves.png]



- [aml_project/data/processed/gmm_clusters_visualization.png]



- [aml_project/data/processed/gmm_uncertainty_pca.png]